

Effective Properties of Two-Scale Viscoelastic Composites



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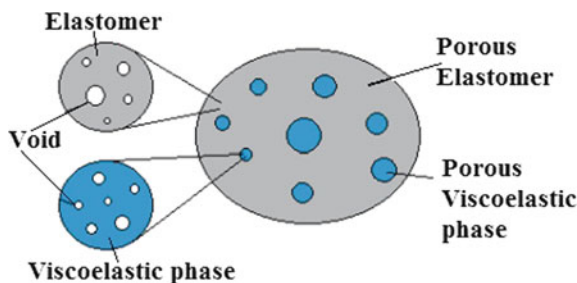
1 Introduction

Polymers commonly exhibit viscoelasticity. They have dissipation effects (from viscous part) as well as storage effects (from elastic part). Viscoelastic composites have gained interest among researchers because of their applications in energy absorbers, dampers, acoustic insulation, thermal insulation, etc. Many works have been carried out to evaluate the effective properties of viscoelastic composites. Eshelby through his seminal paper [1] introduced a new arena of research in the field of composite materials, Effective Behavior of Composite Materials. Thereafter, the area was pioneered by Hill [2, 3], Walpole [4], Mori and Tanaka [5], Hashin and Shtrikman [6], Christensen [7], Mura and Nemat-Nasser [8], etc. For elastic composites, micromechanical methods are very well used to predict the effective properties if the properties of components and the volume fractions of phases are known. Eshelby's approach for elastic materials has been extended by researchers to viscoelastic materials by Laplace–Carson transform or correspondence principle. In this method, the equations for local and effective properties of linear viscoelastic composite after being transformed to Laplace domain can be treated similar to those for elastic composite with complex moduli. Hashin [9] for the very first time considered the problem of spherical viscoelastic inclusion in elastic and viscoelastic matrices. Laws and McLaughlin [10] extended self-consistent method for the estimation of effective properties of (i) sphere-dispersed viscoelastic composite and (ii) fiber-reinforced viscoelastic composite. Brinson and Lin [11] presented a comparison of Mori-Tanaka and finite element unit cell method for viscoelastic composites. Frequency and volume fraction were found to affect the results for different methods. Yi et al. [12] had applied asymptotic homogenization technique for estimating effective moduli of composite with periodic microstructure. Friebel et al. [13] used the correspondence principle to extend the two-step homogenization procedure to estimate the effective properties of

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Fig. 1 Schematic of the microstructure considered



multi-phase viscoelastic composites. Nguyen et al. [14] modeled a porous medium as a mixture of viscoelastic matrix with pore inclusion. Both the matrix and inclusion were modeled as Maxwell elements and the results were in agreement with the exact solution. Lahellec and Suquet [15] developed a direct method based on time-integration approach to estimate the effective properties of composite comprising of linear viscoelastic constituents. Daridon et al. [16] estimated the effective properties of a composite which consisted of matrix, inclusion, and the interface. All of the constituents were modeled as Kelvin–Voigt elements but the resulting effective behavior obtained was of Kelvin–Voigt type with an additional memory term.

In the present article, mean-field-homogenization-based effective properties for viscoelastic composite are estimated. The composite consists of elastomeric matrix and inclusions in the form of pores and porous viscoelastic media (Fig. 1). This paper presents a micromechanical approach to estimating the effective viscoelastic properties of the composite that consists of (i) pores and (ii) porous viscoelastic inclusion.

2 Governing Equations

Consider a material with N different phases that are distributed either randomly or periodically, in the specimen occupying a volume Ω_0 with the boundary $\partial\Omega_0$ such that the characteristic length of inclusion is much smaller than the specimen than length scale separation is used to derive macroscopic properties of the composite using the microstructural properties. In this work, porous viscoelastic composite consisting of pores and porous viscoelastic inclusion is considered. The size of pores in matrix and viscoelastic inclusion is assumed to be much smaller than the size of the viscoelastic inclusion. A representative volume element Ω is considered at the boundary $\partial\Omega$ of which macroscopic strain is related to displacement as $\mathbf{u} = \bar{\mathbf{E}}\mathbf{x}$.

For a linear viscoelastic non-aging material, the stress evolution is related to strain rate as

$$\sigma_{ij}(x, t) = \int_0^t C_{ijkl}(x, t - t') \frac{\partial \varepsilon_{kl}(x, t')}{\partial t'} dt', \quad \forall x \in \Omega \quad (1)$$

where C_{ijkl} is the relaxation modulus. In case of small displacement and neglecting body force, the compatibility condition and equilibrium equation are as follows [17]:

$$\varepsilon_{ij} = \frac{1}{2} [u_{i,j} + u_{j,i}], \quad \forall x \in \Omega \quad (2)$$

$$\sigma_{ij,j} = 0, \quad \forall x \in \Omega \quad (3)$$

For viscoelastic material, the stress–strain relations are converted to linear form by applying Laplace–Carson (LC) transform. Thereafter, the linear viscoelastic problem becomes similar to a linear elastic problem. The viscoelastic properties of the composite are estimated in the LC space by relating the local behavior

$$\sigma^*(x) = \mathbb{C}^*(x) : \varepsilon^*(x) \quad (4)$$

and macroscopic behavior of an RVE

$$\Sigma^*(x) = \mathbb{C}_{hom}^* : E^* \quad (5)$$

In the above equations, (*) superscript terms represent LC transform of the corresponding quantities as

$$\varphi^*(x, p) = p \int_0^\infty \varphi(x, t) e^{-pt} dt \quad (6)$$

The local and average strain tensors are related by the equation

$$\varepsilon^*(x) = A^*(x) : E \quad (7)$$

where $A^*(x)$ is the strain localization tensor at point x . Strain localization tensor for an inclusion embedded in the matrix is obtained by solving Eqs. (1)–(3) along with uniform strain boundary condition. In elastic domain, the problem is solved by displacement approach. Perfect interface is considered between the matrix and inclusion and both the phases are considered to be isotropic. The shear moduli (μ) and bulk moduli (K) are estimated by prony series

$$\mu(t) = \mu_0 \left[1 - \sum_{i=1}^m w_i \left(1 - e^{-\frac{t}{\xi_i}} \right) \right] \quad (8)$$

$$K(t) = K_0 \left[1 - \sum_{i=1}^m w_i \left(1 - e^{-\frac{t}{\xi_i}} \right) \right] \quad (9)$$

Table 1 Mechanical properties of the idealized viscoelastic materials used by Friebe et al. [13]

Shear		Bulk	
τ_i [s]	$\mu_0 w_i$ [bar]	τ_i [s]	$K_0 w_i$ [bar]
0.032	2.512	100.000	3000
0.100	10.000	316.228	100
0.316	56.234		
1.000	316.228		
3.162	1000.000		
10.000	199.526		
31.623	50.119		
100.000	19.953		
316.228	12.589		
1000.000	2.512		
3162.278	1.698		
10,000.000	1.202		
31,622.777	1.148		
100,000.000	1.096		
	$\mu_0 = 1677.979$		$K_0 = 3300$

where μ_0 and K_0 are instantaneous shear and bulk moduli, and $w_i K_0$ and $w_i \mu_0$ are weights corresponding to relaxation time ξ_i . The representative values for prony series have been taken from Friebe et al. [13] and are listed in Table 1.

The relaxation tensors for the phases are given as

$$\mathbb{C}^{i*} = 3K_i^* \mathbb{J} + 2\mu_i^* \mathbb{K} \quad (10)$$

$$\mathbb{J}_{ijkl} = \frac{1}{3} \delta_{ij} \delta_{kl}, \quad \mathbb{K}_{ijkl} = \frac{1}{2} \left(\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk} - \frac{2}{3} \delta_{ij} \delta_{kl} \right) \quad (11)$$

where K_i^* is the bulk relaxation and G_i^* is the shear relaxation. For particulate composites, the global strain localization tensor is related to local strain localization tensor as

$$\mathbb{A}^* = A^{i*} \left(c_0 I + \sum_{r=1}^N c_r A^{r*} \right)^{-1} = A^{i*} \left(\sum_{r=0}^N c_r A^{r*} \right)^{-1} \quad (12)$$

where c_0 is the volume fraction of the matrix, c_r is the volume fraction of different inclusions, and A^{i*} is the local strain localization tensor. Also, $A^{0*} = I$ has been used. For any phase(s),

$$\mathbb{A}_s^{i*} = [I + \mathbb{P}_s^{i*} (C^s - C^0)]^{-1} \quad (13)$$

where the polarization tensor $\mathbb{P}$ for a spherical inclusion in a matrix with shear and bulk moduli μ_0, K_0 is given by

$$\begin{aligned}\mathbb{P}^0 &= \frac{1}{3K_{p0}}\mathbb{J} + \frac{1}{2\mu_{p0}}\mathbb{K} \\ 3K_{p0} &= 3K_0 + 4\mu_0, \quad 2\mu_{p0} = \frac{5\mu_0}{3} \frac{3K_0 + 4\mu_0}{K_0 + 2\mu_0}\end{aligned}\quad (14)$$

The Mori–Tanaka scheme when applied to matrix-inclusion composite considers the matrix to be reference medium [19]. The subscript 0 denotes matrix whereas the subscripts 1, 2, ..., N denote different inclusions. Effective modulus of the composite obtained using Mori–Tanaka model is

$$\mathbb{C}_{hom}^* = \sum_{r=0}^N c_r \mathbb{C}^r \mathbb{A}_r^i \left(\sum_{s=0}^N c_s \mathbb{A}_s^i \right)^{-1} \quad (15)$$

The individual behavior of porous elastomer and porous viscoelastic phase is given by

$$\begin{aligned}\bar{K}_1^* &= \frac{4}{3} \frac{1-p_1}{p_1} \mu_1^{0*}, & \bar{\mu}_1^* &= \frac{3(1-p_1)}{3+2p_1} \mu_1^{0*} \\ \bar{K}_2^* &= \frac{4\mu_2^{0*} K_2^{0*} (1-p_2)}{4\mu_2^{0*} + 3p_2 K_2^{0*}}, & \bar{\mu}_2^* &= \frac{(9K_2^{0*} + 8\mu_2^{0*})(1-p_2)\mu_2^{0*}}{K_2^{0*}(9+6p_2) + \mu_2^{0*}(8+12p_2)}\end{aligned}\quad (16)$$

where p_1 and p_2 represent porosity in elastomer and viscoelastic phase, respectively, also (μ_1^{0*}, K_1^{0*}) and (μ_2^{0*}, K_2^{0*}) are shear and bulk moduli of elastomer and viscoelastic phase in Laplace–Carson space.

Composite Sphere Assemblage (CSA) [20] model states that the composite can be modeled with the assembly of concentric composite spheres of different sizes. The composite sphere consists of an inner sphere—inclusion and an annular sphere matrix. The size of composite sphere is such that in each sphere the inclusion volume fraction is same as that of the composite. The radius ratio of a (internal sphere) and b (outer sphere) is related to inclusion volume fraction in composite as $(a/b)^3 = f$, f being the volume fraction of inclusion in composite. CSA model gives an exact solution only for bulk modulus as

$$\bar{K}/K = \frac{(1-p)(2-4\nu)}{(2-4\nu) + (1+\nu)p} \quad (17)$$

where p is porosity and ν is Poisson's ratio.

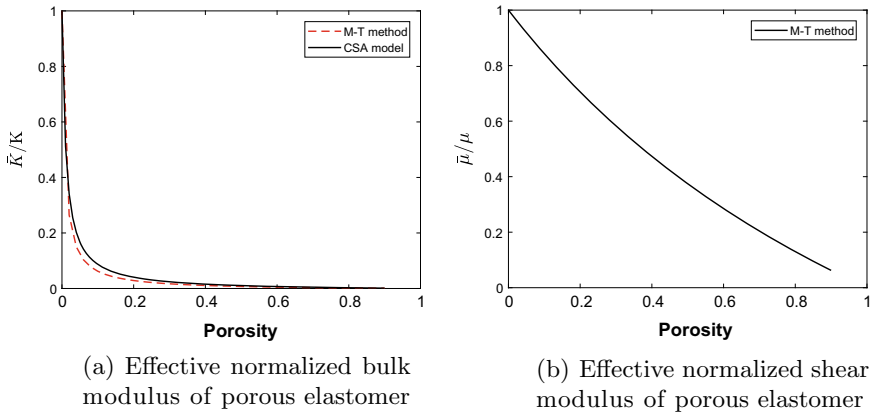


Fig. 2 Effective normalized modulus of porous elastomer

3 Results

In this section, mean-field estimate for macroscopic properties is presented. Silicone rubber is used as matrix material. For silicone rubber, bulk modulus (K) is 1.8 GPa and shear modulus (μ) is 0.01 GPa. By substituting Laplace variable (s) with $i\omega$ we can get frequency-dependent behavior. Here we have taken the frequency to be 1 Hz and the results below are for that. Porosity in the matrix as well in inclusion varies from 0 to 70%. The effective bulk modulus ($\bar{K}$) is observed to decay exponentially with the variation of porosity (Fig. 2a). $\bar{K}$ drops to $\sim 7\%$ at 20% porosity which indicates the sensitivity of $\bar{K}$ (i.e., resistance to uniform compression) to porosity. However, in case of shear modulus, the behavior is different as it decreases gradually with porosity (Fig. 2b).

Mechanical loss factor or dissipation factor ($\tan \delta$) is measure of material's ability to dissipate energy in addition to storage of energy. Higher mechanical loss factor ensures better dissipation of energy. For a composite with only porous inclusions and no porosity in matrix, $\tan \delta$ is observed to increase up to 0.85 volume fraction of viscoelastic inclusion and thereafter it decreases but with the increase in volume fraction of viscoelastic inclusion $\tan \delta$ keeps increasing (Fig. 3). Viscoelastic phase thus helps in easy energy dissipation. For the second case, when non-porous inclusions are embedded in porous matrix, it is observed that dissipation is better at high volume fraction of inclusion but matrix porosity has adverse effect on its dissipation property (Fig. 4).

Fig. 3 Effect of inclusion pore fraction on $\tan \delta$ of the composite

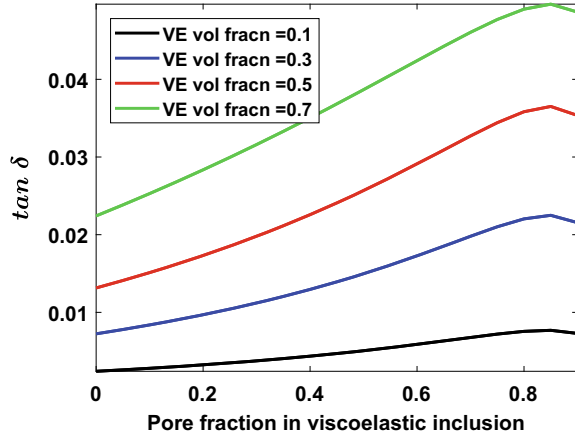
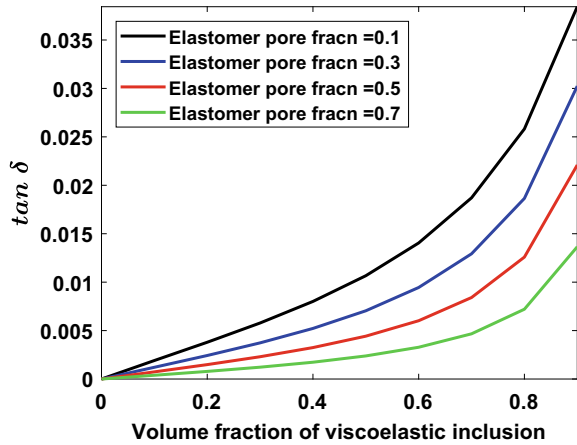


Fig. 4 Effect of viscoelastic phase volume fraction on $\tan \delta$ of the composite



3.1 Effect of Porosity on Effective Behavior

In this section, we present the results about the effect of matrix and inclusion porosity on effective properties of the composite. Figure 5 shows the results for three cases of porosity in inclusion: (i) 0.3, (ii) 0.5, (iii) 0.7.

Bulk modulus has an exponential decay with the addition of porous inclusion as well as matrix porosity (Fig. 5a). The composite's effective bulk modulus $\bar{K}$ is sensitive to porosity and inclusion. Young's modulus and shear modulus are almost similar in nature with a gradual growth (Fig. 5b, c). The growth of shear modulus and hence Young's modulus is due to the fact that shear modulus for viscoelastic phase is much higher than that of silicone rubber. Also it can be seen that addition of porous inclusion reduces the normalized modulus of composite. Addition of porous inclusions enhances the ability of composite to dissipate energy whereas the addition

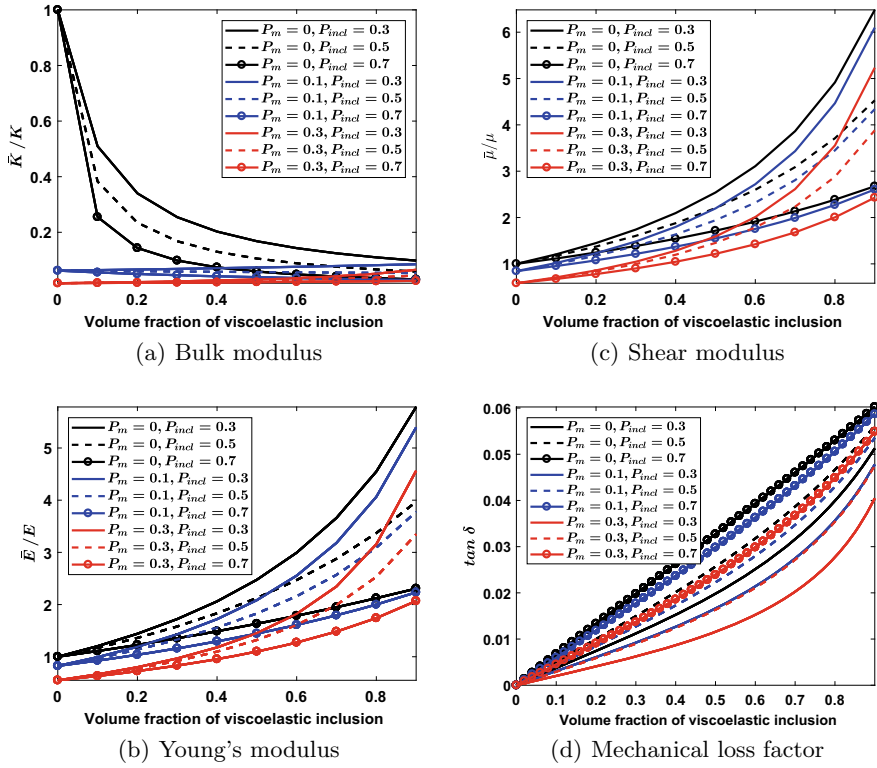


Fig. 5 Effective normalized elastic modulus of porous composite

of porosity in matrix is seen to weaken the energy dissipation behavior of composite. Also the increase of viscoelastic phase also improves the composite's ability to dissipate energy.

4 Conclusion

The present study for two-scale viscoelastic composites was carried out in linear elasticity and viscoelasticity. Linear viscoelastic problem was converted to linear elastic one using Laplace–Carson transform. The effective behavior of heterogeneous material can be determined by mean-field estimation technique (Mori–Tanaka method). Here Mori–Tanaka method was used to estimate effective properties of the composite.

1. The microstructure of the composite is governed by the porosity in the constituent phases and their volume fractions.

2. The dissipation effect in the composite is affected by the microstructure of the composite, i.e., the volume fraction of inclusion and porosity in the inclusion and matrix.
3. Bulk modulus has an exponential decay whereas shear modulus has a gradual decay with an increase in matrix porosity, and addition of viscoelastic inclusion improves Shear and Young's modulus.
4. With the addition of porous inclusions dissipation effect gets enhanced.

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